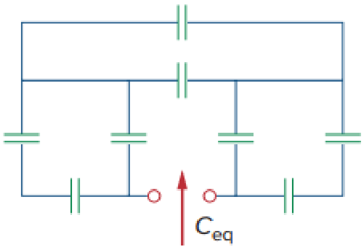
Chapter 6, Problem 18P

Problem

Find Ceq in the circuit of Fig. 6.52 if all capacitors are 4 μF.

Fig. 6.52:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

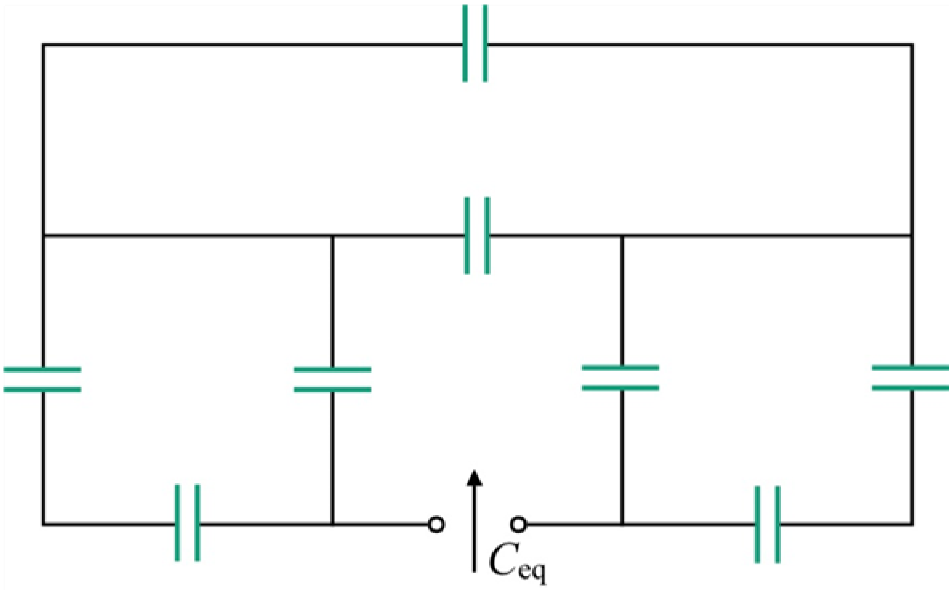


Figure 1

Step 2 of 6

Consider the following data:

The each value of capacitor shown in Figure 1 is $4\ \mu\text{F}$

Now draw the circuit with capacitance values.

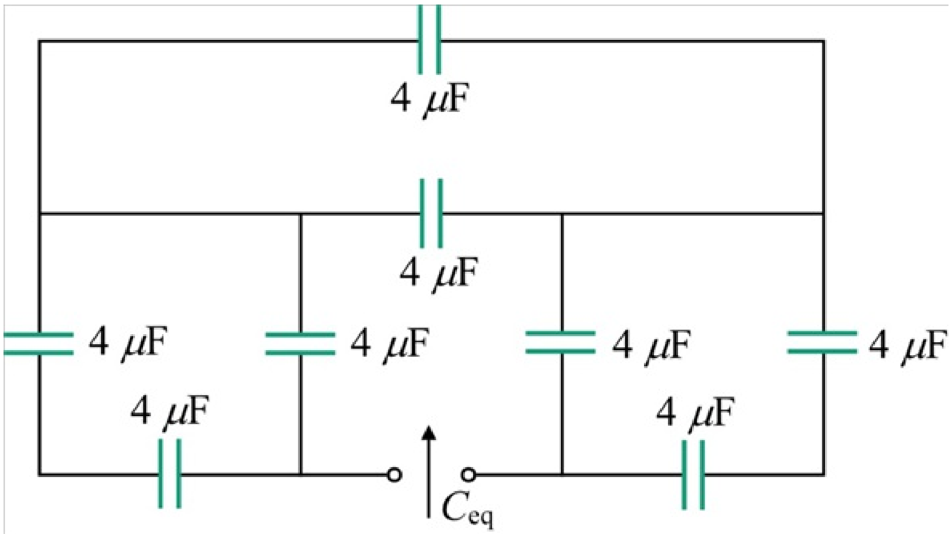


Figure 2

Step 3 of 6

In the Figure 2, two $4\ \mu\text{F}$ capacitors are connected in series. Calculate the equivalent capacitance of series connected capacitors and assign as C_{eq1}.

$$\begin{aligned} C_{\text{eq1}} &= \frac{(4\ \mu\text{F})(4\ \mu\text{F})}{4\ \mu\text{F} + 4\ \mu\text{F}} \\ &= \frac{16\ \mu\text{F}}{8} \\ &= 2\ \mu\text{F} \end{aligned}$$

Step 4 of 6

Draw the modified circuit after converting series capacitors into equivalent capacitance.

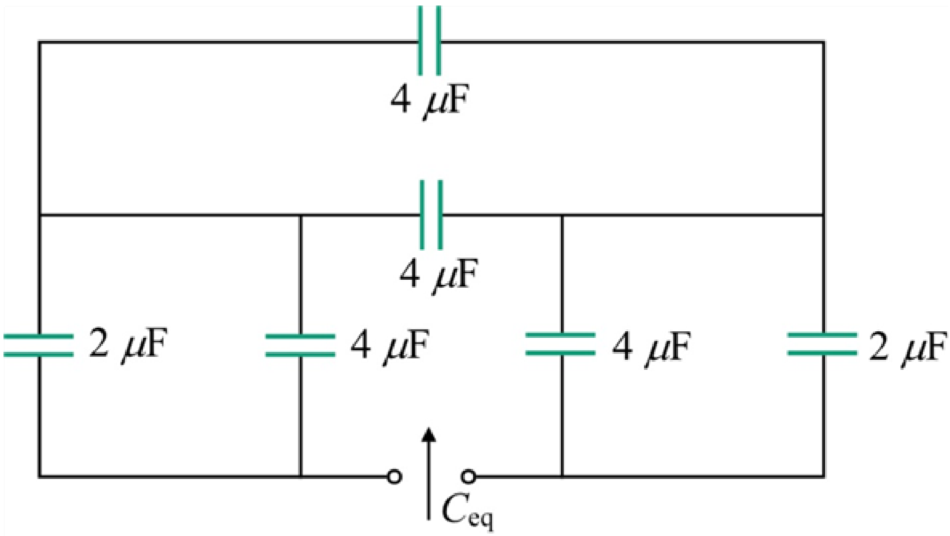


Figure 3

Step 5 of 6

In the Figure 2, $2\ \mu\text{F}$ and $4\ \mu\text{F}$ capacitors are connected in parallel. Calculate the equivalent capacitance of parallel connected capacitors and assign as C_{eq2}.

$$\begin{aligned} C_{\text{eq2}} &= 2\ \mu\text{F} + 4\ \mu\text{F} \\ &= 6\ \mu\text{F} \end{aligned}$$

Similarly, two $4\ \mu\text{F}$ capacitors are connected in parallel. Calculate the equivalent capacitance of parallel connected capacitors and assign as C_{eq3}.

$$\begin{aligned} C_{\text{eq3}} &= 4\ \mu\text{F} + 4\ \mu\text{F} \\ &= 8\ \mu\text{F} \end{aligned}$$

Step 6 of 6

Draw the modified circuit diagram, after converting parallel capacitors into equivalent.

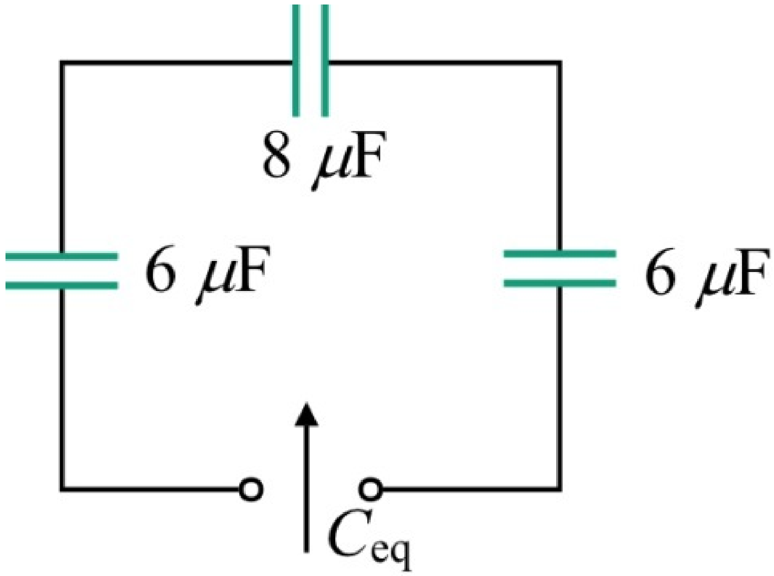


Figure 4

In the Figure 4, two $6\ \mu\text{F}$ and one $8\ \mu\text{F}$ capacitors are connected in series. Calculate the equivalent capacitance of the series connected capacitors.

$$\begin{aligned} C_{\text{eq}} &= \frac{1}{\frac{1}{6\ \mu\text{F}} + \frac{1}{8\ \mu\text{F}} + \frac{1}{6\ \mu\text{F}}} \\ &= \frac{1}{\frac{8+6+8}{48\ \mu\text{F}}} \\ &= \frac{48\ \mu\text{F}}{22} \\ &= 2.1818\ \mu\text{F} \end{aligned}$$

Therefore, the equivalent capacitance, C_{eq} is $\boxed{2.1818\ \mu\text{F}}$.